

KRISHNENDU MAJUMDER

AI/ML Developer | Data Analytics | Remote Sensing | Computer Vision

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PROFILE / PROFESSIONAL SUMMARY

Computer Science graduate with hands-on experience in AI/ML, data analytics, remote sensing, and computer vision. Currently working as an AI/ML Developer Intern at PlanetEye Farm-AI on satellite-based crop classification and infrastructure-monitoring workflows. Skilled in Python, machine learning, exploratory data analysis, feature engineering, model evaluation, data visualization, Google Earth Engine, and Sentinel-1/2 imagery.

PROJECTS

1. Olist Brazilian E-Commerce Data Analytics

Analyzed ~100,000 Brazilian e-commerce orders (2016-2018) using Python, Pandas, NumPy, SQL/SQLite, Matplotlib, and Seaborn. Evaluated revenue, AOV, sales trends, customer geography, delivery performance, product categories, and ratings; identified ~R\$15.42M revenue, ~R\$159.85 AOV, and a rating decline from 4.29 to 2.27 for late deliveries.

2. Satellite-Based Crop Classification & Prediction

Built a multi-temporal workflow using Sentinel-1/2 bands, vegetation indices, temporal features, and ML classifiers to distinguish crop categories across growth stages.

3. Transmission Tower Detection & Thermal Distress Monitoring

Developed an infrastructure-monitoring workflow using RGB imagery and object detection, with thermal/LST and nearby vegetation analysis to support tower-condition assessment.

4. P-Median Facility Location Optimization - IBM CPLEX (OPL)

Implemented a Mixed Integer Programming model for a nine-city network, assigning customers to facilities while selecting P=3 warehouse locations under service constraints.

TECHNICAL SKILLS

Programming: Python, Java, C

Data Analytics: Pandas, NumPy, Data Cleaning, EDA, KPI Analysis, Customer/Sales Analysis, Statistical Analysis

Machine Learning: Classification, Random Forest, XGBoost, LightGBM, CatBoost, Extra Trees, SVM, Feature Engineering, Model Evaluation, Threshold Tuning

Computer Vision: YOLO, Object Detection, Image Preprocessing, Dataset Preparation, Satellite Image Enhancement

Data Visualization: Matplotlib, Seaborn, Geographic/Map-based Visualization

Databases: SQL, SQLite

Geospatial & Tools: Google Earth Engine, Sentinel-1/2, KML/GIS, Git, IBM CPLEX / OPL

PROFESSIONAL EXPERIENCE

AI/ML Developer Intern - PlanetEye Farm-AI Limited

Apr 2026 - Oct 2026

6-Month Paid Internship

- Working on multi-temporal crop classification and prediction using Sentinel-1 SAR, Sentinel-2 multispectral imagery, vegetation indices, and Google Earth Engine.
- Using geospatial/KML field data for satellite feature extraction, ML workflows, model evaluation, and classification threshold tuning.
- Contributing to transmission-tower monitoring using object detection, satellite-image enhancement, thermal/LST analysis, and vegetation/NDVI context.

EDUCATION

B.Tech - Computer Science Engineering, RCC Institute of Information Technology, Kolkata

2022 - 2025

Diploma - Computer Science & Technology, Elite Institute of Engineering & Management, Kolkata

2019 - 2022

Higher Secondary - HARE SCHOOL, Kolkata

2019

Secondary Education - HARE SCHOOL, Kolkata

2017

LANGUAGES

English | Hindi | Bengali

CERTIFICATIONS / TRAINING

Full Stack Development in Python - National Institute for Industrial Training, Kolkata

Jul-Aug 2024